

A NOTE ON ε -DISTANCE BETWEEN MACHINES

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This note proves that the state construction extends to PMet -enriched categories.

THE STATE CONSTRUCTION

Consider a copy-discard category $\mathcal{C}$, and the feedback copy-discard category of machines on it, $\text{St}_{\mathcal{C}}$. Explicitly, $\text{St}_{\mathcal{C}}$ has the same objects as $\mathcal{C}$ and its morphisms of type $A \rightarrow B$ are pairs $(f|S)$ of an object S of $\mathcal{C}$ - the state space - and a morphism $f: S \otimes A \rightarrow S \otimes B$ in $\mathcal{C}$ - the transition morphism - quotiented by bisimilarity: the equivalence relation generated by $(f|S) \sim (g|T)$ if there is a deterministic and total morphism $u: S \rightarrow T$ in $\mathcal{C}$ such that $\text{--}f \text{--}u \text{--} = \text{--}u \text{--}g \text{--}$.

REMARK

We could consider machines with initial states without much effort. For simplicity, we disregard them for now.

Consider the monoidal category $(\text{PMet}, \oplus, 1)$ of pseudometric spaces and nonexpansive functions with monoidal product given by $(X, d_x) \oplus (Y, d_y) := (X \times Y, d_x + d_y)$.

The category PMet has another monoidal structure $(X, d_x) * (Y, d_y) := (X \times Y, \max(d_x, d_y))$.

Together, these two monoidal structures make PMet a duoidal category because, for all $w, x, y, z \in [0, \infty)$,

$$\max(w+x, y+z) \leq \max(w, y) + \max(x, z).$$

Duoidal categories are natural bases of enrichment for monoidal (and copy-discard) categories.

Suppose that we are given a PMet -enriched copy-discard category $\mathcal{C}$. We want to lift this enrichment to $\text{St}_{\mathcal{C}}$. We do this by defining ε -bisimilarity first, mimicking the coalgebraic definition of ε -bisimilarity and ε -distance.

ε -MORPHISMS OF MACHINES

An ε -morphism $(u, \varepsilon): (f|S) \rightarrow (g|T)$ between machines is a deterministic and total morphism $u: S \rightarrow T$ in $\mathcal{C}$ such that $d(\underline{f-u}, \underline{u-g}) \leq \varepsilon$.

ε -BISIMULATION BETWEEN MACHINES

An ε -bisimulation between two machines, $(f|S)$ and $(g|T)$, is a zig-zag of ε_i -morphisms

$$(f|S) \xrightarrow{(u_1, \varepsilon_1)} (h_1|R_1) \xleftarrow{(u_2, \varepsilon_2)} (h_2|R_2) \xrightarrow{(u_3, \varepsilon_3)} \dots \xrightarrow{(u_m, \varepsilon_m)} (h_m|R_m) \xrightarrow{(u_{m+1}, \varepsilon_{m+1})} (g|T)$$

such that $\sum \varepsilon_i \leq \varepsilon$.

Two machines are ε -bisimilar, $(f|S) \approx_\varepsilon (g|T)$, if there is an ε -bisimulation between them.

DISTANCE BETWEEN MACHINES

The distance between machines is the smallest ε for which they are ε -bisimilar:

$$d((f|S), (g|T)) := \inf \{ \varepsilon \in [0, \infty] \mid (f|S) \approx_\varepsilon (g|T) \}.$$

REMARK

This construction is really lifting the enrichment of $\mathcal{C}$ to an enrichment of $\text{St}_{\mathcal{C}}$ as a monoidal bicategory. When we look at ε -bisimilarity and ε -distance, we are looking at the shadow of the bicategorical structure on the first dimension.

EXAMPLE

We may consider $\mathcal{C} = \text{Kl } \mathcal{D}$ with the PMet -enrichment defined by the Wasserstein distance. Then, the ε -bisimilarity and ε -distance that we obtain coincide with the coalgebraic notions.

CLAIM 1

Machines over a PMet-enriched symmetric monoidal category $\mathcal{C}$ form a monoidal bicategory where a morphism $(u, \varepsilon): (f|S) \rightarrow (f'|S')$ is a morphism $u: S \rightarrow S'$ in $\mathcal{C}$ with $\varepsilon \in \mathbb{R}_{\geq 0}$ such that $d(\overline{f-u}, \overline{-u-f'}) \leq \varepsilon$.

PROOF SKETCH

The structure of 1-cells is the usual state construction; the coherence diagrams for 2-cells commute as for the usual state construction.

We check that composition and monoidal product of 2-cells is well-defined.

Suppose that $(u, \delta): (f|S) \rightarrow (f'|S')$ and $(v, \varepsilon): (g|T) \rightarrow (g'|T')$. Then $(u \otimes v, \delta + \varepsilon): (f|S); (g|T) \rightarrow (f'|S'); (g'|T')$ because

$$\begin{aligned} & d(\overline{\overline{f} \overline{g} \overline{u} \overline{v}}, \overline{\overline{u} \overline{f'} \overline{v} \overline{g'}}) \\ & \text{(interchange)} \\ & = d(\overline{\overline{f-u} \overline{g-v}}, \overline{\overline{u-f'} \overline{v-g'}}) \\ & \text{(\mathcal{C} enriched)} \\ & \leq d(\overline{f-u}, \overline{-u-f'}) + d(\overline{g-v}, \overline{-v-g'}) \\ & \text{(def. of morphisms)} \\ & = \delta + \varepsilon \end{aligned}$$

Similarly, $(u \otimes v, \max\{\delta, \varepsilon\}): (f|S) \otimes (g|T) \rightarrow (f'|S') \otimes (g'|T')$ because

$$\begin{aligned} & d(\overline{\overline{f} \overline{g} \overline{u} \overline{v}}, \overline{\overline{u} \overline{f'} \overline{v} \overline{g'}}) \\ & \text{(interch.)} \\ & = d(\overline{\overline{f-u} \overline{g-v}}, \overline{\overline{u-f'} \overline{v-g'}}) \\ & \text{(\mathcal{C} enriched)} \\ & \leq \max\{d(\overline{f-u}, \overline{-u-f'}), d(\overline{g-v}, \overline{-v-g'})\} \\ & \text{(def. of morphs)} \\ & = \max\{\delta, \varepsilon\} \end{aligned}$$

□

For convenience, let us write $\delta \wedge \varepsilon := \max\{\delta, \varepsilon\}$

CLAIM 2

- PMet-enriched copy-discard category
- ⇒ St_ε PMet-enriched with the distance above

PROOF

By definition, the distance between machines is reflexive, symmetric and satisfies the triangle inequality.

We check that composition and monoidal product are nonexpansive.

We need to prove that

$$\begin{aligned} & \bullet d((f|S);(g|T), (f'|S'); (g'|T')) \\ & \leq d((f|S), (f'|S')) + d((g|T), (g'|T')) \\ & \bullet d((f|S) \otimes (g|T), (f'|S') \otimes (g'|T')) \\ & \leq d((f|S), (f'|S')) \wedge d((g|T), (g'|T')) \end{aligned}$$

We prove a stronger statement:

if $(f|S) \approx_{\delta} (f'|S')$ and $(g|T) \approx_{\varepsilon} (g'|T')$ then

$$\begin{cases} (f|S);(g|T) \approx_{\delta+\varepsilon} (f'|S');(g'|T') \\ (f|S) \otimes (g|T) \approx_{\delta \wedge \varepsilon} (f'|S') \otimes (g'|T') \end{cases}$$

We prove this by induction on the length n of the zig-zags witnessing the δ -bisimilarity $(f|S) \approx_{\delta} (f'|S')$ and the ε -bisimilarity $(g|T) \approx_{\varepsilon} (g'|T')$.

For the base case $n=0$, we have

$$(id, \delta): (f|S) \rightarrow (f'|S') \text{ and } (id, \varepsilon): (g|T) \rightarrow (g'|T').$$

By claim 1, we obtain that

$$\begin{cases} (id, \delta+\varepsilon): (f|S);(g|T) \rightarrow (f'|S');(g'|T') \\ (id, \delta \wedge \varepsilon): (f|S) \otimes (g|T) \rightarrow (f'|S') \otimes (g'|T') \end{cases}$$

which give zig-zags of length 0 witnessing

$$\begin{cases} (f|S);(g|T) \approx_{\delta+\varepsilon} (f'|S');(g'|T') \\ (f|S) \otimes (g|T) \approx_{\delta \wedge \varepsilon} (f'|S') \otimes (g'|T') \end{cases}$$

For the inductive step, suppose that the statement holds for length n and that

$$\begin{aligned} (f|S) & \approx_{\delta_1} (f_1|S_1) \xleftarrow{(u_2, \delta_2)} (f_2|S_2) \xrightarrow{(u_3, \delta_3)} (f'|S') \\ (g|T) & \approx_{\varepsilon_1} (g_1|T_1) \xleftarrow{(v_2, \varepsilon_2)} (g_2|T_2) \xrightarrow{(v_3, \varepsilon_3)} (g'|T') \end{aligned}$$

with $\approx_{\delta_1}$ and $\approx_{\varepsilon_1}$ of length at most n , and $\delta_1 + \delta_2 + \delta_3 \leq \delta$ and $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 \leq \varepsilon$.

We know that

$$\begin{cases} (u_2, \delta_2): (f_2|S_2) \rightarrow (f_1|S_1) \\ (u_3, \delta_3): (f_2|S_2) \rightarrow (f'|S') \\ (v_2, \varepsilon_2): (g_2|T_2) \rightarrow (g_1|T_1) \\ (v_3, \varepsilon_3): (g_2|T_2) \rightarrow (g'|T') \end{cases}$$

⇒ by claim 1, we obtain

$$\begin{cases} (u_2 \otimes v_2, \delta_2 + \varepsilon_2): (f_2|S_2);(g_2|T_2) \rightarrow (f_1|S_1);(g_1|T_1) \\ (u_3 \otimes v_3, \delta_3 + \varepsilon_3): (f_2|S_2);(g_2|T_2) \rightarrow (f'|S');(g'|T') \\ (u_2 \otimes v_2, \delta_2 \wedge \varepsilon_2): (f_2|S_2) \otimes (g_2|T_2) \rightarrow (f_1|S_1) \otimes (g_1|T_1) \\ (u_3 \otimes v_3, \delta_3 \wedge \varepsilon_3): (f_2|S_2) \otimes (g_2|T_2) \rightarrow (f'|S') \otimes (g'|T') \end{cases}$$

By induction, there are zig-zags

$$\begin{cases} (f|S);(g|T) \approx_{\delta_1 + \varepsilon_1} (f_1|S_1);(g_1|T_1) \\ (f|S) \otimes (g|T) \approx_{\delta_1 \wedge \varepsilon_1} (f_1|S_1) \otimes (g_1|T_1) \end{cases}$$

⇒ we obtain zig-zags

$$\begin{cases} (f|S);(g|T) \approx_{\delta+\varepsilon} (f'|S');(g'|T') \\ (f|S) \otimes (g|T) \approx_{\delta \wedge \varepsilon} (f'|S') \otimes (g'|T') \end{cases}$$

because

$$\begin{aligned} & \delta_1 + \varepsilon_1 + \delta_2 + \varepsilon_2 + \delta_3 + \varepsilon_3 = \delta_1 + \delta_2 + \delta_3 + \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \leq \delta + \varepsilon \\ & \delta_1 \wedge \varepsilon_1 \wedge \delta_2 \wedge \varepsilon_2 \wedge \delta_3 \wedge \varepsilon_3 = \delta_1 \wedge \delta_2 \wedge \delta_3 \wedge \varepsilon_1 \wedge \varepsilon_2 \wedge \varepsilon_3 \leq \delta \wedge \varepsilon \end{aligned}$$

By definition of distance between machines, we obtain that, if $d((f|S), (f'|S')) = \delta$ and $d((g|T), (g'|T')) = \varepsilon$ then there are zig-zags witnessing that $(f|S) \approx_{\delta} (f'|S')$ and $(g|T) \approx_{\varepsilon} (g'|T')$.

By the inductive proof, we obtain zig-zags witnessing

$$\begin{cases} (f|S);(g|T) \approx_{\delta+\varepsilon} (f'|S');(g'|T') \\ (f|S) \otimes (g|T) \approx_{\delta \wedge \varepsilon} (f'|S') \otimes (g'|T') \end{cases}$$

⇒ by definition of distance, we obtain that

$$\begin{aligned} d((f|S);(g|T), (f'|S');(g'|T')) & \leq \delta + \varepsilon \\ & = d((f|S), (f'|S')) + d((g|T), (g'|T')) \end{aligned}$$

and

$$\begin{aligned} d((f|S) \otimes (g|T), (f'|S') \otimes (g'|T')) & \leq \delta \wedge \varepsilon \\ & = d((f|S), (f'|S')) \wedge d((g|T), (g'|T')) \end{aligned}$$

□